

SLIIT - IT4010 - Research Project

Checklist 1 Submission

Digital Platform for Social Welfare Services Management in Sri Lanka

Research Project : Group 25-26J-320

Specialization: Information Technology (IT)

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Repository Link

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1. Project Overview

This research project aims to address systemic inefficiencies in Sri Lanka's current social welfare ecosystem, which primarily relies on manual application processing and fragmented data management systems.

To overcome these challenges, the proposed solution is a **centralized, AI-powered digital platform** designed for citizens, Divisional Secretariat officers, and policymakers. The platform enhances transparency, accessibility, and efficiency in welfare service delivery.

The system leverages **predictive analytics and machine learning techniques** to ensure that benefits are allocated to genuinely deserving citizens. Additionally, it provides proactive, data-driven decision-making tools to support government authorities in improving welfare distribution.

2. System Architecture

The proposed system follows a **multi-layered architecture**, consisting of the following components:

- **User Interface Layer**
Provides interaction interfaces for citizens and government officers.
- **AI Intelligence Layer**
Handles data processing, prediction models, and intelligent decision-making.
- **Infrastructure Layer**
Ensures secure data storage, system integration, and API management.

Core System Components

1. Eligibility & Healthcare Engine

- **Description:** Predicts citizen eligibility for welfare programs and recommends appropriate healthcare specialists.
- **Novelty:** Provides AI-driven personalized assistance based on individual citizen profiles.

2. Multilingual Chatbot

- **Description:** Offers real-time assistance in **Sinhala, Tamil, and English**.
- **Novelty:** Introduces a tri-lingual chatbot tailored for Sri Lankan welfare services.

3. Financial Guidance Module

- **Description:** Recommends savings plans and micro-investment opportunities for low-income beneficiaries.
- **Novelty:** Promotes financial empowerment through intelligent recommendations.

4. AI + GIS Mapping Module

- **Description:** Identifies regional patterns and hotspots of vulnerable populations.
- **Novelty:** Enables predictive geo-mapping for targeted and efficient resource allocation.

3. Technical Dependencies

Programming Languages

- Python (AI/ML development)
- JavaScript (Web interface development)

AI & NLP Frameworks

- Rasa
- spaCy

Machine Learning Techniques

- Decision Trees
- Logistic Regression
- Clustering Algorithms

Geospatial Technologies

- GIS Mapping Systems

Infrastructure

- Web Server
- Secure APIs
- Centralized Database System

Project Summary

This project focuses on addressing inefficiencies in Sri Lanka's existing social welfare system, which is currently hindered by manual processes and fragmented data management.

The proposed AI-powered platform introduces a centralized system that improves transparency, accessibility, and efficiency for all stakeholders, including citizens, government officers, and policymakers.

By integrating predictive analytics and machine learning, the system ensures fair and accurate welfare distribution while enabling data-driven governance and decision-making.